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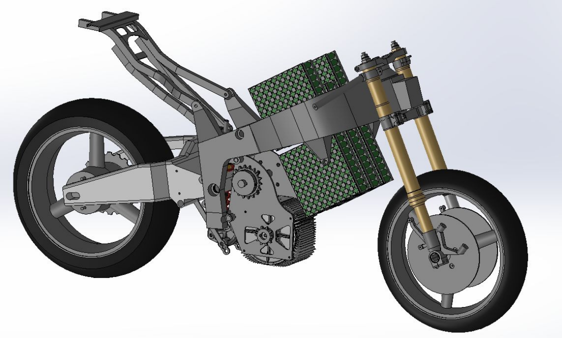
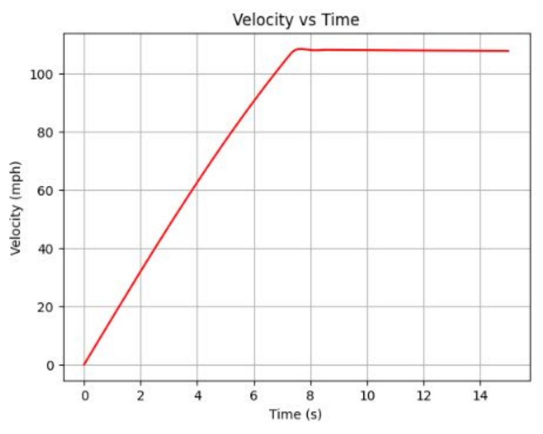
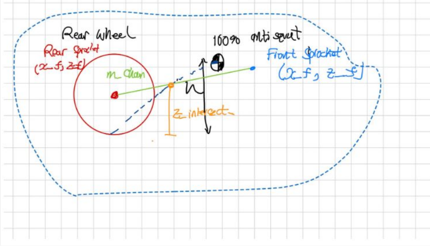


Ghost Electric Motorcycles - Drivetrain (Design)

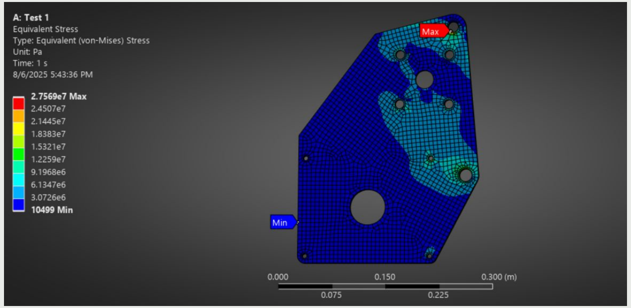


Velocity vs Time derived from a Python based physics sim

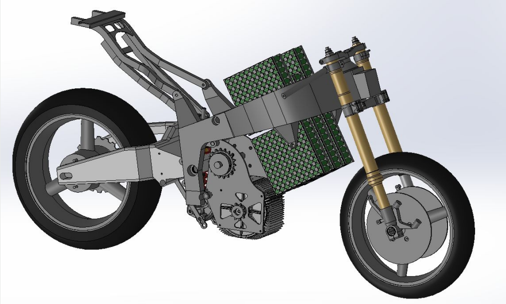



```python
[ ] def antisquat(x_f, z_f, x_r, z_r, h_cg):

        #x_f, z_f : coordinates of the front sprocket
        #x_r, z_r : coordinates of the rear sprocket (or axle)
        #h_cg     : center of gravity height


        # Calculate slope of chain line
        m_chain = (z_f - z_r) / (x_f - x_r)

        # Chain line height at rear axle vertical
        z_intersect = m_chain * (x_r - x_f) + z_f

        # Anti-squat percentage
        antisquat_percent = (z_intersect / h_cg) * 100

        return antisquat_percent, z_intersect

        Returns:
        antisquat_percent : Anti-squat percentage
        z_intersect       : height where chain line crosses vertical at rear axle
```

Anti-squat calculator



FEA for motor mount



Final layout

What:

-Designed and optimized the drivetrain system for Ghost's first electric motorcycle
-Built on the frame of a '05 Kawasaki ZX-6R

How:

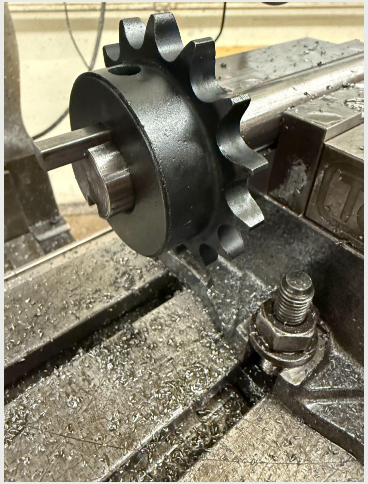
- Determined ideal gear ratio (5.2 : 1) and jackshaft placement through Python based physics simulations
-Designed drivetrain assembly in SolidWorks that achieved desired characteristics
-Free body diagram and FEA to understand the forces and stresses within components

Results:

-Presented design in a CDR to engineers from Harley-Davidson, Lucid Motors, and LiveWire.
-Received "go-ahead" for manufacturing

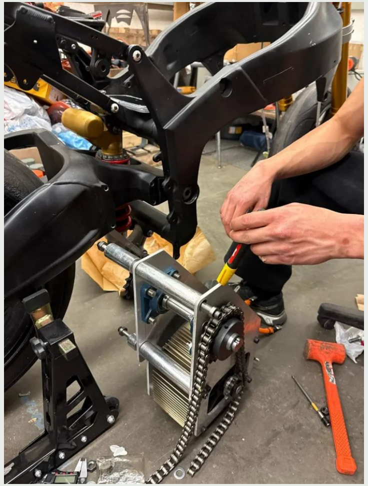
More granular information can be found here:
charlesxiao.xyz/ghost

Ghost Electric Motorcycles - Drivetrain (Build and Test)



Jackshaft milling

Structural beam fit up



Drivetrain assembly

Final Assembly

First rolling test

<u>What:</u>

I successfully built and tested the drivetrain system I designed

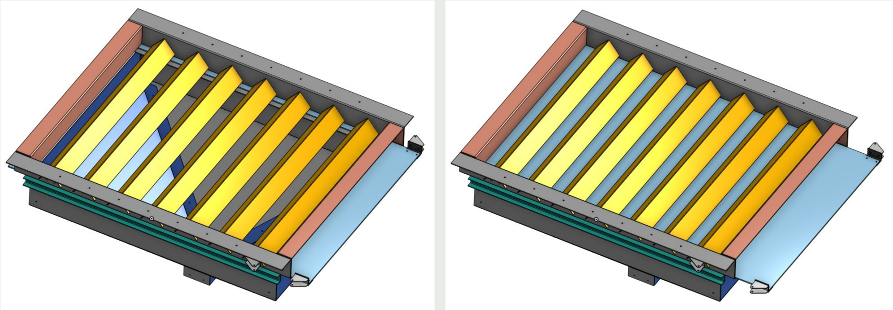
<u>How:</u>

-Machined 6 unique components on manual mill and manual lathe within 0.005”
-Waterjet motor mounts - ¼” A36 steel

<u>Results:</u>

-<$300 for stock material and COTS components
-It worked!

Terraform Industries - Reactor Outlet



Outlet gate in open and closed positions

Picture of the full reactor - base outlined

First prototype - Served as proof of concept

Second prototype installed under reactor

Outlet actuation in motion (double click the image)

What:

Designed and built a structural base and outlet for a 1400 kg carbonation reactor.

How:

-Ran tests to determine the minimum wall angle to facilitate mass flow with sorbent flakes

-Designed components in Onshape sheet metal

-Performed stress calculations to size sheet metal thickness with a FoS of 1.5

-Sized linear actuators to open and close the gate

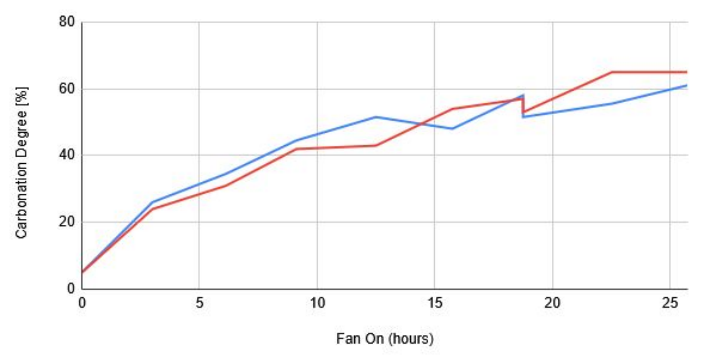
Results:

-<15 minute build time

-Achieved stable mass flow in the reactor

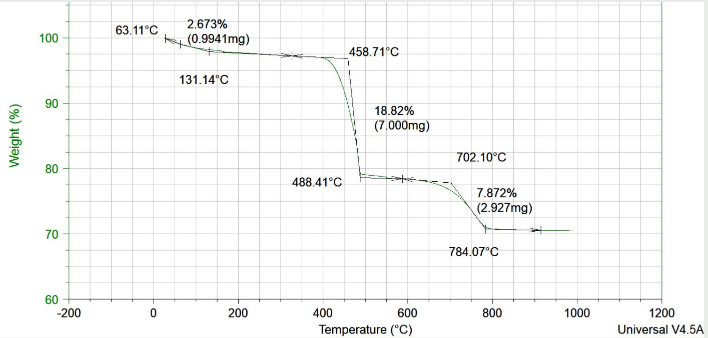
-Next prototype will be deployed at Terraform's first test site in the Mojave desert

Terraform Industries - Carbonation Test Campaign

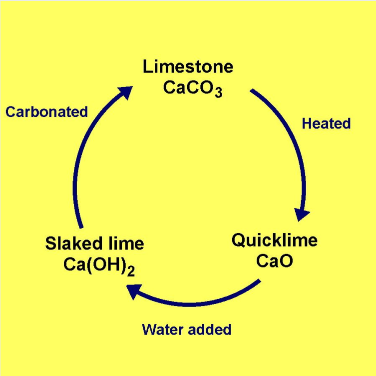


Carbonation Percentage vs Time

Calcium hydroxide sorbent flakes



Graph showing the mass drops from the TGA



Carbonation reactor works reacting:
$Ca(OH)2 + CO2 = CaCO3 + H2O$

What:

Developed and ran a test campaign to characterize the rate of CO2 capture as a function of time

How:

-Collected samples from multiple locations within the reactor at varying time intervals.
-Ran samples through a thermogravimetric analyzer (TGA) and evaluated mass change to determine carbonation percentage
-Analyzed power consumption and CO2 capture rates to optimize throughput rate

Results:

Discovered that reducing the residency time of sorbent material from 24 hours to 15 hours reduced reactor power consumption by 13% while capturing the same amount of CO2

FPV Racing Drone (Personal Project)

What:

Built an FPV drone because I saw clips of them on YouTube and thought it was the coolest thing ever

How:

Selected and purchased each component and assembled them myself

Results:

It went really fast! - 120 km/h. You can find clips of it on my website charlesxiao.xyz/drone.html